

Accelerated Article Preview

Evaluating large language models for accuracy incentivizes hallucinations

Received: 1 July 2025

Accepted: 15 April 2026

Accelerated Article Preview

Published online: 22 April 2026

Cite this article as: Kalai, A. T. et al. Evaluating large language models for accuracy incentivizes hallucinations. *Nature* <https://doi.org/10.1038/s41586-026-10549-w> (2026)

Adam Tauman Kalai, Ofir Nachum, Santosh S. Vempala & Edwin Zhang

This is a PDF file of a peer-reviewed paper that has been accepted for publication. Although unedited, the content has been subjected to preliminary formatting. Nature is providing this early version of the typeset paper as a service to our authors and readers. The text and figures will undergo copyediting and a proof review before the paper is published in its final form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers apply.

Evaluating large language models for accuracy incentivizes hallucinations

Adam Tauman Kalai^{1,*}, Ofir Nachum¹, Santosh S. Vempala², Edwin Zhang^{1,†}

¹ OpenAI, San Francisco, CA, USA.

² Georgia Institute of Technology, Atlanta, GA, USA.

† Present address: Isara Laboratories, San Francisco, CA, USA.

* Correspondence: Adam Tauman Kalai, adam@kal.ai

Abstract

Large language models sometimes produce confident, plausible falsehoods (“hallucinations”), limiting their reliability^{1,2}. Prior work has offered numerous explanations and effective mitigations such as retrieval and tool use³, consistency-based self-verification⁴, and reinforcement learning from human feedback⁵. Nonetheless, the problem persists even in state-of-the-art language models^{6,7}. Here we show how next-word prediction and accuracy-based evaluations inadvertently reward unwarranted guessing. Initially, next-word pretraining creates statistical pressure toward hallucination even with idealized error-free data: using learning theory^{8,9}, we show that facts lacking repeated support in training data (such as one-off details) yield unavoidable errors, while recurring regularities (such as grammar) do not. Subsequent training stages aim to correct such errors. However, dominant headline metrics like accuracy systematically reward guessing over admitting uncertainty. To align incentives, we suggest two additions to the classic approach of adding error penalties to evaluations to control abstention^{10,11}. First, we propose “open-rubric” evaluations that explicitly state how errors are penalized (if at all), which test whether a model modulates its abstentions to stated stakes while optimizing accuracy. Second, since hallucination-specific benchmarks rarely make leaderboards¹², we suggest using open-rubric variants of existing evaluations, to reverse their guessing incentives. Reframing hallucination as an incentive problem opens a practical path toward more reliable language models.

Plausible, confidently stated falsehoods diminish the utility of large language models (LLMs) in reliability-critical domains. Despite progress, this problem persists even in state-of-the-art models^{6,7}. For example, three popular LLMs (details in Methods) responded to the question “*What does PGGB stand for?*” as follows:

ChatGPT: “*PGGB*” can stand for different things depending on context, but one of the most common meanings is “**Polynomial Gaussian Gradient Bandwidth**”...

Claude: *PGGB* most commonly stands for **Privately Held Global Growth and Income Fund**, a closed-end fund that trades on the NYSE...

DeepSeek: Of course! **PGGB** stands for **Perfect Guard Group Buy**...

All are wrong. Why not abstain, for example by saying “I don’t know” or by requesting more context, as a reliable human assistant would? Instead, they fabricate specific, confident responses. Like students facing a difficult exam question, models guess rather than admit uncertainty. To reduce hallucinations further, we need to understand why they arise and persist (Fig. 1).

How do hallucinations arise? Various explanations have been offered¹, and we provide a unifying lens using computational learning theory⁸. To make this lens precise, we first address a practical obstacle: defining and measuring hallucination is complicated by issues such as responses that contain multiple vague claims¹³. We sidestep these definitional tangles by analyzing an abstract set of errors, following learning theory, which applies to binary classification (e.g., dog versus cat images) without adjudicating edge cases (e.g., images containing both).

Initially, LLMs are trained to optimize next-word (or next-token) prediction, during *pretraining*. While falsehoods in training data can cause hallucinations, the phenomenon is not purely garbage-in garbage-out. We show that this objective creates statistical tendency toward hallucination even with ideal error-free training data. The key insight is a formal reduction to binary classification: any language model implicitly answers “*Is this response valid?*” for each candidate generation. This connection establishes lower bounds on hallucination rates and illuminates which error types to expect, since causes of misclassification are well understood⁹. Learnable patterns (grammar, spelling, politeness) yield low error rates, while non-learnable facts (birthdays, one-off details) produce high hallucination rates in pretrained LLMs. A natural conjecture is that removing errors and including abstentions in the training corpus would resolve most hallucinations, but our analysis shows this is not the case.

Why do hallucinations persist, despite the fact that numerous alignment approaches have been shown to curb them²? We offer a simple explanation: under standard scoring, guessing is a dominant strategy. In terms of raw accuracy or pass rate (fraction of overall questions answered correctly), a model that guesses outscore a trustworthy model that abstains when uncertain. For example, on the SimpleQA evaluation¹⁴, accuracy slightly favors OpenAI’s o4-mini which answers almost all questions (with over 3/4 error rate)¹⁵ over GPT-5-mini, even though GPT-5-mini makes many fewer errors (due to abstentions)⁶. Our meta-evaluation of popular LLM benchmarks confirms that the majority favor guessing. The ubiquity of rewarding guessing explains why the handful of existing hallucination evaluations also have not resolved the problem¹². Instead, we argue that existing primary evaluations need to be revised. One may hope that accuracy will approach 100% through innovations such as web search and scaling. However, some real-world questions will remain unanswerable (e.g., unlisted birthdays or impossible questions), so accuracy-style metrics still favor guessing when uncertain.

Finally, how can evaluation incentives be realigned to reduce hallucination? A key challenge is that the true response utility is context-specific and subjective. OpenAI¹⁶ specifies only a vague ranking: correct answer > no answer > wrong answer. If numeric scores are not aligned with utility, overconfident or underconfident models will outscore the ideal LLM which appropriately expresses uncertainty.

We propose a straightforward method to modify existing questions to make scoring unambiguous: state the scoring system alongside the question itself. Such “open rubric” questions can be paired with the classic approach of encouraging abstention by penalizing errors^{10,11}. For example, for the recent “net score”⁷ prompts could be augmented with “correct answers receive 1 and incorrect answers receive −1 (hence abstain if < 50% likely to be correct).” When there is no penalty, “only fully correct answers receive credit (so make your best guess if unsure)” describes standard accuracy.

Under open rubrics, accuracy is no longer at odds with humility: a reliable model can guess when instructed to do so and abstain when not, just as people guess more on exams than in everyday settings. The GPT-5-mini model mentioned above which often abstained, now becomes more accurate than o4-mini when measured with open rubrics (Methods). A model which performs well across penalties demonstrates controllable abstention. We provide a case study using four frontier models, demonstrating how open rubrics incentivize adoption of a hallucination-reduction technique. Taken together, our analysis reframes hallucination as an unintended outcome of training objectives and evaluation incentives rather than an inherent LLM deficiency, and demonstrates how aligned evaluations can incentivize reliability in language models.

Hallucination from next-word pretraining

Our analysis offers an organizing lens for a range of error phenomena, including hallucination, that arise when generating with calibrated next-word predictors. In what statistical sense does hallucination differ from misspelling? Pretraining fits a probability model $\hat{p}$ to text sampled from a distribution p using maximum-likelihood estimation, as in autocomplete systems. With ideal training data, autocomplete should generate correct spelling, yet it may still fabricate dates when completing “Adam Tauman Kalai was born on...” if that birthday is absent from the training data. (This is only expected from pretraining, before further alignment steps.) The challenge in formalizing this intuition is that falsehoods would not be generated by the perfect predictor $\hat{p} = p$ trained on infinite error-free training data; nor would they be generated by the untrained LLM which completes everything with “I don’t know.” Our analysis applies to large but finite error-free datasets which may contain ample abstentions. With errors in training data, one may expect a higher hallucination rate.

We show that generating valid outputs, for calibrated LLMs, is harder than classifying output validity. Specifically, consider an Is-It-Valid (IIV) binary classification problem which has a training set consisting of a large number of responses, each labeled either as valid (+) or error (−), as illustrated in Fig. 2. For this supervised learning problem, both train and test data are 50/50 mixtures of valid examples labeled as + (i.e., the pretraining data since we assume it is valid) and uniformly random errors labeled as −. We then show how any LLM can be used as an IIV classifier. Theorem 1 relates IIV and generative errors (such as hallucinations):

$$(\text{generative error rate}) \gtrsim 2 \cdot (\text{IIV misclassification rate}).$$

Well-understood error decompositions in supervised learning⁹ thus translate to generative errors. Fig. 2 (right) illustrates two causes visually: middle, a poor model of a linear separator for a circular region (i.e., approximation error due to hypothesis-class misspecification); and bottom, statistical complexity (i.e., estimation error due to finite samples and capacity control).

Statistical complexity is a common cause of hallucination arising when there is no succinct pattern that fully describes the concept. Knowledge gaps are inevitable with non-exhaustive training data, e.g., one cannot predict birthdays absent from the training data. Using the IIV reduction, Theorem 3 (Methods) shows a stylized setting where

$$(\text{hallucination rate}) \gtrsim (\text{fraction of training facts that appear exactly once}).$$

For instance, if 20% of birthday facts appear exactly once in pretraining data, then pretrained models should hallucinate on at least 20% of birthday facts. On the other hand, LLMs rarely

hallucinate on country capitals, which all appear repeatedly in training data. Theorem 3 extends prior work in this stylized setting¹⁷ which did not consider prompts or abstention.

We now present our main reduction without prompts. The analysis with prompts (Theorem 2 in Methods) is similar. Without prompts, a pretrained LLM $\hat{p}$ is a probability distribution over a set $\mathcal{X}$ of “plausible” texts x , e.g., statements or documents. We assume that $\mathcal{X}$ is finite for simplicity. The examples $\mathcal{X} = \mathcal{E} \cup \mathcal{V}$ are partitioned into errors $\mathcal{E}$ and valid examples $\mathcal{V}$, for nonempty disjoint sets $\mathcal{E}, \mathcal{V}$. For hallucinations, $\mathcal{E}$ would consist of plausible generations containing one or more falsehoods. The error rate of LLM $\hat{p}$ is

$$\text{err} := \hat{p}(\mathcal{E}) = \Pr_{x \sim \hat{p}}[x \in \mathcal{E}].$$

Pretraining data are assumed to come from a noiseless *pretraining distribution* p over $\mathcal{X}$, i.e., $p(\mathcal{E}) = 0$.

The IIV binary-classification problem is formally specified by the target function $f : \mathcal{X} \rightarrow \{-, +\}$ to be learned (membership in $\mathcal{V}$) and the distribution D over examples $\mathcal{X}$ (a 50/50 mix of samples from p and uniformly random errors):

$$D(x) := \begin{cases} p(x)/2 & \text{if } x \in \mathcal{V}, \\ 1/(2|\mathcal{E}|) & \text{if } x \in \mathcal{E}, \end{cases} \text{ and } f(x) := \begin{cases} + & \text{if } x \in \mathcal{V}, \\ - & \text{if } x \in \mathcal{E}. \end{cases}$$

The LLM is thus used as an IIV classifier, in our reduction, by thresholding its probability $\hat{p}(x)$ (which can generally be efficiently computed) at $1/|\mathcal{E}|$. The IIV *misclassification rate* is

$$\text{err}_{\text{iiv}} := \Pr_{x \sim D}[\hat{f}(x) \neq f(x)], \text{ where } \hat{f}(x) := \begin{cases} + & \text{if } \hat{p}(x) > 1/|\mathcal{E}|, \\ - & \text{if } \hat{p}(x) \leq 1/|\mathcal{E}|. \end{cases}$$

Theorem 1. *For any pretraining distribution p such that $p(\mathcal{E}) = 0$ and any LLM $\hat{p}$,*

$$\text{err} \geq 2 \cdot \text{err}_{\text{iiv}} - \frac{|\mathcal{V}|}{|\mathcal{E}|} - \delta,$$

where $\delta := |\hat{p}(\mathcal{T}) - p(\mathcal{T})|$ and $\mathcal{T} := \{x \in \mathcal{X} \mid \hat{p}(x) > 1/|\mathcal{E}|\}$.

Since Theorem 1 holds for *any* LLM $\hat{p}$, it immediately implies that all pretrained LLMs will err on inherently unlearnable IIV facts (such as birthdays absent from the training data) where err_{iiv} is necessarily large, and where δ and $|\mathcal{V}|/|\mathcal{E}|$ are small (e.g., for each person there are 364 more incorrect birthday claims in $\mathcal{E}$ than correct ones in $\mathcal{V}$, though $\mathcal{V}$ also includes valid abstentions).

Since pretraining aims for $\hat{p} \approx p$, one may expect small $\delta = |\hat{p}(\mathcal{T}) - p(\mathcal{T})|$ which is a weak form of what is known as *calibration*. Formally, it has been shown that maximum likelihood estimation (i.e., minimizing cross-entropy) leads to calibration¹⁸, and pretrained LLMs empirically satisfy related forms of calibration^{19,20}. However, aligned models should not hallucinate and hence should be poorly calibrated. The Methods section contains proofs and shows how other types of hallucination are related to misclassification.

Evaluation metrics that reward guessing

Beyond next-word prediction, a number of techniques have demonstrated significant empirical reductions in hallucination rates², some of which show substantial gains even on frontier models. Yet

these methods are typically judged (and models are selected) using headline metrics like accuracy or pass rate, which treat abstention as failure and thus reward guessing. Here we observe that the metrics used in existing benchmarks and leaderboards reinforce hallucination when a model is unsure.

Most leaderboards score each problem (or each subpart) as either correct or incorrect, and abstention is typically graded as incorrect. Such binary evaluations impose a false right-wrong dichotomy, awarding no credit to answers that appropriately express uncertainty, omit dubious details, or request clarification. Under this scoring, abstaining is strictly suboptimal, being penalized as incorrect while an overconfident “best guess” is optimal for maximizing expected accuracy. As a result, accuracy-based evaluation pushes models to convert uncertainty into content that is often plausible but wrong. Observation 1 (Methods) formally encapsulates how guessing is a dominant strategy for binary evaluations. Reducing hallucination often means lower accuracy, which impedes adoption.

Table 1 (Extended Data) summarizes our meta-evaluation (Methods), finding that the vast majority of popular evaluations use binary grading. Therefore, additional hallucination evaluations may not suffice when the primary evaluations penalize honestly reporting uncertainty. This does not diminish existing work on hallucination evaluations but rather highlights that even an ideal one may still be outweighed by lower scores on the vast majority of existing benchmarks.

Scoring metrics do not merely measure model performance: in practice, scores are effectively optimized throughout LLM development, including the selection of data, architectures, and algorithms. As a result, accuracy-centric leaderboards can perpetuate hallucination by rewarding guessing when a model is unsure, motivating evaluation designs that break this cycle.

Aligning evaluation incentives

Because evaluations inform which LLMs are deployed, their scoring shapes model behavior. A classic approach to discouraging guessing is to penalize errors^{10,11}. More generally, for each response r to a question prompt (or context) $c \in \mathcal{C}$, we seek a score $s(c, r) \in \mathbb{R}$ that approximates response utility $u(c, r)$. Utility is context-specific: confabulated acronyms are less costly than hallucinations about elevator capacity limits, and overestimates on a capacity sign are worse than small mistakes helping students study. The recent “net score”⁷ assigns 1 to correct answers and -1 to incorrect answers. This incentivizes guessing only when the probability of being correct exceeds 50%, which is not uniformly appropriate (though arguably superior to the status quo 0% threshold).

One way to make scoring unambiguous is to explicitly specify the scoring rubric in the prompt, removing the need to subjectively assess error severity. We refer to a question as open-rubric (OR) if it is clear to the test-taker how errors and abstentions are scored. Sample instruction templates used in our case study are in Extended Data Table 2.

Error penalties have traditionally been applied at the scoring stage and are therefore invisible to the model. A model that is told the cost of errors can adjust abstention to the stakes, much as students adjust guessing on exams to a known grading system. An error penalty $L \geq 0$ is strategically equivalent to offering a reward of $t := L/(1 + L)$ for abstaining. While error penalties are common in human exams, we use abstention rewards here because incentives are simpler to understand: if abstaining earns 75% credit, then it is optimal to guess when one is more than 75% likely to be correct. Extended Data Table 2 gives two ways to describe rubrics in text (e.g., explaining that errors cost 3 versus abstaining receives 75% credit). In either case, we refer to t as

the correctness threshold because both views incentivize guessing when one is more than t likely to be correct.

Formally, the abstention reward function assigns scores 1 (correct), t (abstain), and 0 (error):

$$s_t(c, r) := g_c(r) + t \cdot \mathbb{1}[r \in \mathcal{A}_c]$$

where $g_c(r) \in \{0, 1\}$ grades correctness for question c and response r , $\mathcal{A}_c$ is a set of abstention responses, and the indicator $\mathbb{1}[\phi]$ denotes 1 if predicate ϕ holds and 0 otherwise. The corresponding error-penalty score is a rescaling $s'_t(c, r) := (s_t(c, r) - t)/(1 - t)$, with 1 (correct), 0 (abstain), and $-L$ (error) where $L = t/(1 - t)$. Let c_t denote the question augmented with the scoring rubric. For alignment, assuming utility $u(c_t, r) = s_t(c_t, r)$ is more reasonable than $u(c, r) = s_t(c, r)$ for t unspecified in c . At least this rewards a desirable behavior: being correct as often as possible while abstaining when the correctness probability is below the stated threshold. Open rubrics mirror typical human settings: students know whether or not they are taking an exam (and how wrong answers are scored), and guess accordingly. By making scoring visible, open rubrics let LLMs make the same distinction rather than collapsing all contexts into a single mode.

To make these incentives concrete, we run a stylized experiment to see whether open-rubric or closed-rubric evaluations are more likely to favor adopting a hallucination mitigation. The mitigation is tested on frontier reasoning models: Google’s Gemini 3 Pro (LLM1), OpenAI’s GPT-5 (LLM2), xAI’s Grok 4 (LLM3), and Anthropic’s Claude Opus 4.5 (LLM4). This is not a controlled evaluation across models: we use default settings, with no tuning or cost normalization.

We use the SimpleQA evaluation¹⁴ which comprises 4,326 factual questions, such as *What years was Antonio de Padua María Severino López de Santa Anna y Pérez de Lebrón vice president?* Accuracy is often used as a headline metric for SimpleQA²¹ despite its authors’ intentions¹⁴. We begin with the standard closed-rubric (CR) setting, where the scoring rubric is not mentioned in its questions. The baseline condition prompts the LLM to answer the factual question as stated, without additional instruction. We consider a prototypical hallucination mitigation: (a) the model generates two independent responses; (b) those two responses (in addition to the query) are presented to the same model to judge consistency; and (c) if they are judged to be consistent, the first output is selected, otherwise the model abstains. Similar consistency approaches have been studied for detecting and reducing hallucination^{20,22}. The mitigation was chosen for simplicity; many hallucination reduction techniques have been studied, some more effective or efficient². As shown in Fig. 3, the mitigation cuts errors but also reduces accuracy across models.

We next append the rubric strings rub_t from Table 2 to each question with $t = L/(L+1)$, explicitly stating an abstention reward. (The abstention reward version was chosen for simplicity, but a small ablation with error penalties suggested similar performance.) We use thresholds $t = 0, 0.5, 0.75, 0.9$ corresponding to penalties $L = 0, 1, 3, 9$. Models abstain more for larger thresholds t (Extended Data Fig. 2). We again evaluate baseline and mitigation, now prompting with the augmented question. The same mitigation as above is used (including the appended rubric) for $t > 0$. At $t = 0$, however, because the model is instructed to never abstain, a different prompt is used to select the better of the two responses. Conditioning on different error penalties is impossible with closed rubrics because they are not stated.

With an open rubric, the mitigation outperforms the baseline across models and penalties, including zero (accuracy). Open rubrics thus offer consistent encouragement to adopt the mitigation, while closed-rubric accuracy discourages adoption. Open rubrics also provide a natural way to detect overconfidence. For penalties $L = 3, 9$, all four models in Extended Data Table 3 exhibit

strong overconfidence with negative (unmitigated) scores, so always abstaining yields a higher score. For open rubrics with no penalty $L = 0$, one cannot be overconfident because the instructions are to always make one’s best guess.

Discussion

Current LLM development inadvertently limits reliability through multiple stages: pretraining creates statistical pressure toward hallucination, and the prevalence of closed-rubric accuracy-based evaluations means that LLMs are always in exam mode. Improving reliability is thus not only a modeling problem but also an evaluation mechanism-design problem. Rather than adding separate hallucination benchmarks, we argue that primary evaluations should be modified so that they incentivize admitting uncertainty when appropriate. Open rubrics are one such modification: reducing hallucinations no longer comes at the cost of headline metrics. Error penalties and abstention-aware scoring are established ideas¹⁰, but with LLMs the scoring stakes can be stated explicitly. Evaluating across a range of penalties measures a model’s ability to modulate abstention according to the stated penalty—a good student is both knowledgeable and attuned to when guessing is appropriate.

However, closed rubrics have the advantage of ecological validity, as users rarely specify a confidence threshold in real-world prompts. Furthermore, closed-rubric evaluations with appropriate context-specific scores could measure the ability to assess error severity (though assessing different harms is challenging). The correct/incorrect/abstain trichotomy used here is deliberately simple, and it would be desirable to adapt existing evaluations to handle linguistic calibration (“I would guess...”)²³ and the broader pragmatics of language^{24,25}. While hallucination rates have decreased substantially since early models^{6,7}, further progress may depend as much on what we measure as on what we build: aligning evaluation incentives can make reliability improvements pay off.

References

- [1] Huang, L. *et al.* A survey on hallucination in large language models: principles, taxonomy, challenges, and open questions. *ACM Trans. Inf. Syst.* **43**, 42 (2025). URL <https://doi.org/10.1145/3703155>.
- [2] Ji, Z. *et al.* Survey of hallucination in natural language generation. *ACM Computing Surveys* **55**, 248:1–248:38 (2023). URL <https://doi.org/10.1145/3571730>.
- [3] Lewis, P. *et al.* Retrieval-augmented generation for knowledge-intensive nlp tasks. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M. & Lin, H. (eds.) *Advances in Neural Information Processing Systems*, vol. 33, 9459–9474 (Curran Associates, Inc., 2020). URL https://proceedings.neurips.cc/paper_files/paper/2020/file/6b493230205f780e1bc26945df7481e5-Paper.pdf.
- [4] Manakul, P., Liusie, A. & Gales, M. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In Bouamor, H., Pino, J. & Bali, K. (eds.) *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 9004–9017 (Association for Computational Linguistics, Singapore, 2023). URL <https://aclanthology.org/2023.emnlp-main.557/>.

- [5] Ouyang, L. *et al.* Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, vol. 35, 27730–27744 (2022). URL https://papers.nips.cc/paper_files/paper/2022/hash/b1efde53be364a73914f58805a001731-Abstract-Conference.html.
- [6] OpenAI. GPT-5 system card. Tech. Rep., OpenAI (2025). URL <https://cdn.openai.com/gpt-5-system-card.pdf>. Accessed: 2025-09-02.
- [7] Anthropic. Claude Opus 4.6 system card. Tech. Rep., Anthropic PBC (2026). URL <https://www-cdn.anthropic.com/0dd865075ad3132672ee0ab40b05a53f14cf5288.pdf>. System card (PDF). Published 2026-02-06. Accessed 2026-02-23.
- [8] Kearns, M. J., Schapire, R. E. & Sellie, L. M. Toward efficient agnostic learning. *Machine Learning* **17**, 115–141 (1994). URL <http://link.springer.com/article/10.1007/BF00993468>.
- [9] Shalev-Shwartz, S. & Ben-David, S. *Understanding machine learning: from theory to algorithms* (Cambridge University Press, 2014).
- [10] Chow, C. K. On optimum recognition error and reject tradeoff. *IEEE Transactions on Information Theory* **16**, 41–46 (1970).
- [11] Xin, J., Tang, R., Yu, Y. & Lin, J. The art of abstention: selective prediction and error regularization for natural language processing. In Zong, C., Xia, F., Li, W. & Navigli, R. (eds.) *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, 1040–1051 (Association for Computational Linguistics, Online, 2021). URL <https://aclanthology.org/2021.acl-long.84/>.
- [12] Maslej, N. *et al.* Artificial Intelligence Index report 2025. Annual Report, AI Index Steering Committee, Institute for Human-Centered AI, Stanford University, Stanford, CA (2025). URL <https://hai.stanford.edu/ai-index/2025-ai-index-report>. Accessed: 27 Jun 2025.
- [13] van Deemter, K. The pitfalls of defining hallucination. *Computational Linguistics* **50**, 807–816 (2024). URL https://doi.org/10.1162/coli_a_00509.
- [14] Wei, J. *et al.* Measuring short-form factuality in large language models (2024). URL <https://arxiv.org/abs/2411.04368>. arXiv:2411.04368.
- [15] OpenAI. OpenAI o3 and o4-mini system card. <https://openai.com/index/o3-o4-mini-system-card/> (2025). Accessed 8 May 2025.
- [16] OpenAI. OpenAI model spec (version 2025-12-18). <https://model-spec.openai.com/2025-12-18.html> (2025). Accessed: 2025-02-23.
- [17] Kalai, A. T. & Vempala, S. S. Calibrated language models must hallucinate. In *Proceedings of the 56th Annual ACM Symposium on Theory of Computing*, STOC 2024, 160–171 (Association for Computing Machinery, New York, NY, USA, 2024). URL <https://doi.org/10.1145/3618260.3649777>.
- [18] Błasiok, J., Gopalan, P., Hu, L. & Nakkiran, P. When does optimizing a proper loss yield calibration? (2023). URL <https://arxiv.org/abs/2305.18764>. arXiv:2305.18764.

- 281 [19] OpenAI. GPT-4 technical report (2023). URL <https://arxiv.org/abs/2303.08774>.
282 arXiv:2303.08774.
- 283 [20] Kadavath, S. *et al.* Language models (mostly) know what they know (2022). URL <https://arxiv.org/abs/2207.05221>.
284 arXiv:2207.05221.
- 285 [21] Pichai, S., Hassabis, D. & Kavukcuoglu, K. A new era of intelligence with Gemini 3 (2025).
286 URL <https://blog.google/products/gemini/gemini-3/#gemini-3>. Accessed: 2025-12-07.
- 287 [22] Abbasi Yadkori, Y. *et al.* Mitigating LLM hallucinations via conformal abstention (2024). URL
288 <https://arxiv.org/abs/2405.01563>. arXiv:2405.01563.
- 289 [23] Mielke, S. J., Szlam, A., Dinan, E. & Boureau, Y.-L. Reducing conversational agents’ over-
290 confidence through linguistic calibration. *Transactions of the Association for Computational*
291 *Linguistics* **10**, 857–872 (2022). URL <https://aclanthology.org/2022.tacl-1.50/>.
- 292 [24] Grice, H. P. Logic and conversation. In Cole, P. & Morgan, J. L. (eds.) *Syntax and Semantics*,
293 *Vol. 3: Speech Acts*, 41–58 (Academic Press, New York, 1975).
- 294 [25] Ma, B. *et al.* Pragmatics in the era of large language models: a survey on datasets, evaluation,
295 opportunities and challenges. In Che, W., Nabende, J., Shutova, E. & Pilehvar, M. T. (eds.)
296 *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*
297 *(Volume 1: Long Papers)*, 8679–8696 (Association for Computational Linguistics, Vienna,
298 Austria, 2025). URL <https://aclanthology.org/2025.acl-long.425/>.

Table 1: Closed-rubric CR@ L and open-rubric OR@ L mean scores with penalties $L = 0, 1$. When the scoring is unspecified (CR), the mitigation reduces CR@0 (accuracy) for all four LLMs, challenging adoption. With open rubrics, the mitigation helps across the board (for $L = 3, 9$ see Extended Data Table 3). Each cell is an average over $n = 4,326$ questions. All shaded cells represent a statistically significant difference between the mitigation and baseline ($P < 10^{-5}$ using a two-sided paired permutation test).

	CR@0	CR@1	OR@0	OR@1
LLM1	0.71	0.43	0.71	0.46
LLM1*	0.61	0.52	0.73	0.55
LLM2	0.49	0.05	0.50	0.18
LLM2*	0.42	0.28	0.51	0.30
LLM3	0.49	0.02	0.49	0.22
LLM3*	0.36	0.21	0.51	0.26
LLM4	0.37	0.07	0.46	0.14
LLM4*	0.34	0.14	0.47	0.17

* with mitigation

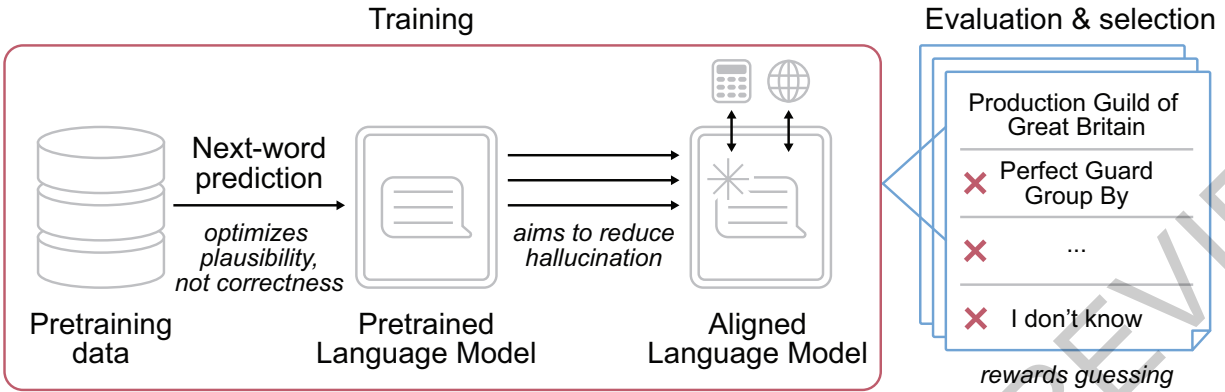


Figure 1: **Origins and persistence of hallucination across training and evaluation.** Next-word prediction creates statistical pressure toward hallucination. Subsequently, numerous techniques improve alignment and reduce hallucination. Yet accuracy-based evaluations ultimately inform model selection, inadvertently incentivizing confident guessing.

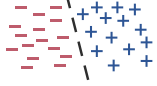
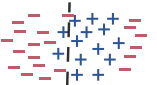
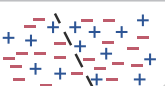
Valid examples +	Error examples -	
Greetings. How can I help?	Greetings. How kan eye help?	 Spelling (good model)
There are 2 D's in LADDER There is 1 N in PIANO	There are 3 L's in SPELL There is 1 G in CAT	 Counting (poor model)
Mia Holdner's birthday is 4/1. I don't know Zdan's birthday.	Colin Merivale's birthday is 8/29. Jago Pere's birthday is 8/21.	 Birthdays (no pattern)

Figure 2: Is-It-Valid requires learning to identify valid generations using labeled $\pm$ examples (left). Classifiers (dashed lines) may be accurate on certain concepts like spelling (top) but errors often arise due to poor models (middle) or arbitrary facts when there is no pattern in the data (bottom).

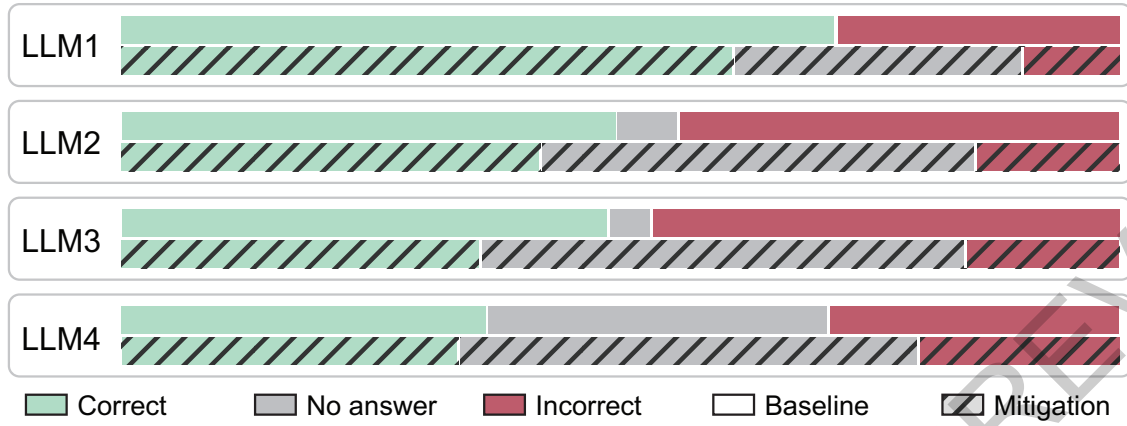


Figure 3: The consistency-based mitigation reduces incorrect responses (hallucination rate) but also decreases correct responses (accuracy), posing a barrier to adoption ($n = 4,326$ questions per model).

Methods

Introductory example details

The prompt “What does PGGB stand for?” was given to GPT-5 (auto), Claude Sonnet 4.5, and DeepSeek (with DeepThink), all accessed 2025-11-11. The models chose to (or did not have access to) search the web.

The SimpleQA (again $n = 4,326$) evaluation of GPT-5-mini and o4-mini is described in the experimental details section below.

Next-word prediction and calibration

When we refer to next-word prediction, we mean factoring a probability of a sequence into individual word probabilities $\hat{p}(w_1 w_2 \dots w_n) = \prod_i \hat{p}(w_i | w_1 \dots w_{i-1})$. This is usually done with “tokens” rather than words, but this distinction is unimportant for our analysis. Using tokens, images may also be represented as sequences.

We now argue why $\delta = |\hat{p}(\mathcal{T}) - p(\mathcal{T})|$ in Theorem 1 is a measure of (mis)calibration that should be small due to pretraining. Note that *without any knowledge of the language*, one can achieve $\delta = 0$ by simply taking the uniform distribution $\hat{p}(x) = 1/|\mathcal{X}|$, and thus $\delta = 0$ does not require $p = \hat{p}$. An auditor can trivially estimate δ by comparing the fractions of responses satisfying $\hat{p}(x) > 1/|\mathcal{E}|$ versus $\hat{p}(\hat{x}) > 1/|\mathcal{E}|$ using sets of training samples $x \sim p$ and synthetic generations $\hat{x} \sim \hat{p}$. Inspired by Dawid²⁶, one may think of an analogy to a weather forecaster predicting the probability of rain each day. A minimal calibration requirement would be whether their average prediction matched the average fraction of rain. One could also require these two to match on days when the forecast was $> t$ for some threshold $t \in [0, 1]$. Dawid²⁶ introduced the more stringent requirement that for *every* $t \in [0, 1]$, among days on which the prediction is t it rains about a t fraction of the time.

Minimizing a variety of losses has been shown to lead to calibration¹⁸. For completeness, here is a particularly simple justification for why δ is typically small for the standard pretraining cross-entropy objective,

$$\mathcal{L}(\hat{p}) = \mathbb{E}_{x \sim p} [-\log \hat{p}(x)]. \quad (1)$$

Consider rescaling the probabilities of the positively-labeled examples by a factor $s > 0$ and normalizing:

$$\hat{p}_s(x) \propto \begin{cases} s \cdot \hat{p}(x) & \text{if } \hat{p}(x) > 1/|\mathcal{E}|, \\ \hat{p}(x) & \text{if } \hat{p}(x) \leq 1/|\mathcal{E}|. \end{cases}$$

Then, a simple calculation shows that δ is the magnitude of the derivative of the loss with respect to the scaling factor s , evaluated at $s = 1$:

$$\delta = \left| \frac{d}{ds} \mathcal{L}(\hat{p}_s) \Big|_{s=1} \right|.$$

If $\delta \neq 0$, then rescaling by some $s \neq 1$ would reduce the loss, so the loss is not at a local minimum. For any class of language models powerful enough to approximate such simple rescaling, local optimization should yield small δ . Note that δ , being defined at a single threshold $t = 1/|\mathcal{E}|$ is weaker than notions such as Expected Calibration Error (ECE) which integrate over thresholds t .

330 The reduction with prompts

331 Theorem 1 will follow from a more general Theorem 2 which covers prompts (contexts) $c \in \mathcal{C}$ drawn
 332 from a *prompt distribution* μ . Henceforth, each example $x = (c, r)$ consists of a prompt c and
 333 plausible response r . Theorem 1 corresponds to the special case in which μ assigns probability 1 to
 334 the empty prompt. For a given prompt $c \in \mathcal{C}$, let $\mathcal{V}_c := \{r \mid (c, r) \in \mathcal{V}\}$ be the valid responses and
 335 $\mathcal{E}_c := \{r \mid (c, r) \in \mathcal{E}\}$ be the erroneous responses. The pretraining distribution and pretrained model
 336 are now conditional response distributions $p(r \mid c), \hat{p}(r \mid c)$. For notational convenience, we extend
 337 these to joint distributions on $\mathcal{X}$ by $p(c, r) := \mu(c)p(r \mid c)$ and $\hat{p}(c, r) := \mu(c)\hat{p}(r \mid c)$, so that still
 338 $\text{err} := \hat{p}(\mathcal{E}) = \sum_{(c,r) \in \mathcal{E}} \mu(c)\hat{p}(r \mid c)$ and $p(\mathcal{E}) = 0$.

339 Pretraining $x_i = (c_i, r_i)$ are like ideal “dialogue” examples, similar to distillation^{27,28}. Although
 340 assuming that the training data contain model dialogues drawn from the same prompt distribution
 341 is unrealistic, even higher error rates may be expected when the assumption fails. The IIV problem
 342 with prompts has the same target function $f(x) := +$ iff $x \in \mathcal{V}$, but the generalized distribution D
 343 selects, with equal probability either $x \sim p$ or $x = (c, r)$ for $c \sim \mu$ and uniformly random $r \in \mathcal{E}_c$.
 344 Finally, the classifier $\hat{f}(c, r)$ is now $+$ iff $\hat{p}(r \mid c) > 1/\min_c |\mathcal{E}_c|$.

Theorem 2. *For any pretraining distribution p such that $p(\mathcal{E}) = 0$ and any pretrained model $\hat{p}$,*

$$\text{err} \geq 2 \cdot \text{err}_{\text{iiv}} - \frac{\max_c |\mathcal{V}_c|}{\min_c |\mathcal{E}_c|} - \delta,$$

345 where $\delta := |\hat{p}(\mathcal{T}) - p(\mathcal{T})|$ for $\mathcal{T} := \{(c, r) \in \mathcal{X} \mid \hat{p}(r \mid c) > 1/\min_c |\mathcal{E}_c|\}$.

346 Generalizing the rescaling $\hat{p}_s(r \mid c)$ (normalizing per prompt, still with single parameter s) again
 347 justifies a small $\delta = \left| \frac{d}{ds} \mathcal{L}(\hat{p}_s) \right|_{s=1}$, now for $\mathcal{L}(\hat{p}) := \sum_{(c,r) \in \mathcal{X}} -\mu(c) \log \hat{p}(r \mid c)$.

348 *Proof of Theorem 2.* Let $K := \min_{c \in \mathcal{C}} |\mathcal{E}_c|$ and $k := \max_{c \in \mathcal{C}} |\mathcal{V}_c|$. Also, recall that $\delta = |\hat{p}(\mathcal{T}) - p(\mathcal{T})|$
 349 which can equivalently be written as $\delta = |p(\mathcal{B}) - \hat{p}(\mathcal{B})|$, where $\mathcal{T}, \mathcal{B}$ denote (top or bottom) responses
 350 that are above and below threshold:

$$\mathcal{T} := \{(c, r) \in \mathcal{X} \mid \hat{p}(r \mid c) > 1/K\} \quad (2)$$

$$\mathcal{B} := \{(c, r) \in \mathcal{X} \mid \hat{p}(r \mid c) \leq 1/K\}. \quad (3)$$

351 Partition the hallucination and misclassification rates into above and below threshold rates:

$$\begin{aligned} \text{err} &= \hat{p}(\mathcal{T} \setminus \mathcal{V}) + \hat{p}(\mathcal{B} \setminus \mathcal{V}) \\ \text{err}_{\text{iiv}} &= D(\mathcal{T} \setminus \mathcal{V}) + D(\mathcal{B} \cap \mathcal{V}). \end{aligned}$$

352 Above the threshold, misclassifications $D(\mathcal{T} \setminus \mathcal{V})$ are the sum of $D(c, r)$ only over $(c, r) \in \mathcal{T}$ such
 353 that $r \in \mathcal{E}_c$ —each contributing $D(c, r) = \mu(c)/2|\mathcal{E}_c| \leq \mu(c)/2K$. But each such misclassification also
 354 contributes $\mu(c)\hat{p}(r \mid c) \geq \mu(c)/K$ to hallucinations above the threshold $\hat{p}(\mathcal{T} \setminus \mathcal{V})$. Hence,

$$\hat{p}(\mathcal{T} \setminus \mathcal{V}) \geq 2D(\mathcal{T} \setminus \mathcal{V})$$

355 Thus, it remains only to show that below the threshold:

$$\hat{p}(\mathcal{B} \setminus \mathcal{V}) \geq 2D(\mathcal{B} \cap \mathcal{V}) - \frac{k}{K} - \delta. \quad (4)$$

By definition, $2D(\mathcal{B} \cap \mathcal{V}) = p(\mathcal{B} \cap \mathcal{V}) = p(\mathcal{B})$. Also, there are $|\mathcal{V}_c| \leq k$ valid responses for each c , each one in $\mathcal{B}$ having $\hat{p}(r \mid c) \leq 1/K$, so $\hat{p}(\mathcal{B} \cap \mathcal{V}) \leq \sum_c \hat{p}(c)k/K = k/K$. Hence,

$$\begin{aligned} 2D(\mathcal{B} \cap \mathcal{V}) - \hat{p}(\mathcal{B} \setminus \mathcal{V}) &= p(\mathcal{B}) - \hat{p}(\mathcal{B} \setminus \mathcal{V}) \\ &= p(\mathcal{B}) - (\hat{p}(\mathcal{B}) - \hat{p}(\mathcal{B} \cap \mathcal{V})) \\ &\leq \delta + \hat{p}(\mathcal{B} \cap \mathcal{V}) \leq \delta + \frac{k}{K}. \end{aligned}$$

This is equivalent to Eq. (4), as needed. $\square$

Regarding plausibility, of course the vast majority of strings are gibberish. Note that the above theorem holds with the modified definitions of nonsensical examples $\mathcal{N}$ with partition $\mathcal{X} = \mathcal{N} \cup \mathcal{E} \cup \mathcal{V}$, $\text{err} := \hat{p}(\mathcal{N} \cup \mathcal{E})$, $D(\mathcal{N}) = 0$, and the assumption that $p(\mathcal{V}) = 1$ rather than $p(\mathcal{E}) = 0$.

Arbitrary-fact hallucinations

Random arbitrary facts are a natural special case of estimation error (with high VC dimension²⁹) when there is no succinct pattern that explains the target function (hence epistemic uncertainty). In particular, this section considers valid responses which are random and independent across prompts. Abstention, denoted by $\perp$, is also considered valid.

Definition 1 (Arbitrary Facts). *The following are fixed: an arbitrary prompt distribution $\mu(c)$, an $\perp$ response and, for each prompt c : a response set $\mathcal{R}_c$ and a probability of answering $\alpha_c \in [0, 1]$. Independently for each c , a single correct answer $a_c \in \mathcal{R}_c$ is chosen uniformly at random. Finally, $p(a_c \mid c) = \alpha_c$ and $p(\perp \mid c) = 1 - \alpha_c$ for each $c \in \mathcal{C}$. Thus $\mathcal{E}_c = \mathcal{R}_c \setminus \{a_c\}$ and $\mathcal{V}_c = \{a_c, \perp\}$.*

It is assumed that there is a single way to write any given fact, which can be done for dates by specifying format. However, we again note that one may expect *even more hallucinations* with multiple ways to state each fact. In the case of fixed-format birthdays, $|\mathcal{E}_c| = 364$ and notable people whose birthdays are discussed often would have high $\mu(c)$. Notable birthdays like Einstein’s appear multiple times, whereas others may only occur once, e.g., in an obituary. LLMs seldom err on frequently referenced facts such as Einstein’s birthday. This model extends prior work¹⁷ to account for abstentions and prompts, which were not considered in that work.

Our lower-bound for hallucinations is based on the fraction of prompts appearing just once in the training data, ignoring abstentions.

Definition 2 (Singleton rate). *A prompt $c \in \mathcal{C}$ is a singleton if it appears exactly once in the N training data $\langle (c^{(i)}, r^{(i)}) \rangle_{i=1}^N$ without abstention, i.e., $|\{i : c^{(i)} = c \wedge r^{(i)} \neq \perp\}| = 1$. Let $\mathcal{S} \subseteq \mathcal{C}$ denote the set of singletons and*

$$\text{sr} = \frac{|\mathcal{S}|}{N}$$

denote the fraction of training singletons.

The singleton rate builds on Alan Turing’s elegant “missing-mass” estimator³⁰, which gauges how much probability is still assigned to outcomes that have not yet appeared in a sample from a distribution. Concretely, Turing’s estimate of the unseen-event probability is the fraction of samples appearing exactly once. Intuitively, singletons act as a proxy for how many more novel outcomes you might encounter in further sampling, so their empirical share becomes the estimate for the entire “missing” portion of the distribution. We now state our bounds for Arbitrary Facts.

Theorem 3 (Arbitrary Facts). *In the Arbitrary Facts model, any algorithm which takes N training samples and outputs $\hat{p}$ satisfies, with probability $\geq 99\%$ over $\vec{a} = \langle a_c \rangle_{c \in \mathcal{C}}$ and the N training examples:*

$$\text{err} \geq \text{sr} - \frac{2}{\min_c |\mathcal{E}_c|} - \frac{35 + 6 \ln N}{\sqrt{N}} - \delta.$$

Moreover, there is an efficient algorithm outputting calibrated $\hat{p}$ ($\delta = 0$) that with probability $\geq 99\%$,

$$\text{err} \leq \text{sr} - \frac{\text{sr}}{\max_c |\mathcal{E}_c| + 1} + \frac{13}{\sqrt{N}}.$$

The proof is in the Supplementary Information. Follow-up empirical work largely corroborates this relationship between hallucinations, the singleton rate, and calibration³¹.

To interpret the terms in the above bound, let us walk through a few settings. As mentioned, most country capitals appear numerous times in training data, and with $\text{sr} = 0$, the bound is uninformative. Indeed, models rarely hallucinate country capitals. Next, consider birthdays and death dates, where $|\mathcal{E}_c| = 364$. For celebrities, these dates would appear numerous times. However, mentions of birthdays and death dates may otherwise be one-offs, e.g., obituaries that are published just once. Many other random facts occurring in town meeting notes or sources that are not repeated may be singletons as well.

Finally, consider publications, such as book or article references. Since a primary goal of publishing is to publicize work, work is often mentioned multiple times, e.g., on a person’s web page, a curriculum vitae, a publication server, presentation announcements, and journal references. Based on this logic, one would expect a low singleton rate for such references, yet they have been a prominent source of hallucinations³². This does not contradict the above lower-bound on hallucination rate, and we next discuss how poor representations could be to blame. A neural network is less reliable than a database of publication titles, populated based on training data. This echoes a well-known cause of misclassification, when a poor model is used to represent a certain class, known as “model misspecification” or “approximation error”⁹.

Errors due to poor models

Misclassifications can also arise when the underlying model is poor because (a) the model family cannot represent the concept well, such as linear separators approximating circular regions, or (b) the model family is sufficiently expressive but the model itself is not a good fit. The letter-counting hallucination in Fig. 2 is an example: LLMs represent text using tokens rather than letters, which is poorly suited for letter counting.

Agnostic Learning⁸ addresses (a) by defining the minimal error rate of any classifier in a given family $\mathcal{G}$ of classifiers $g : \mathcal{X} \rightarrow \{-, +\}$:

$$\text{opt}(\mathcal{G}) := \min_{g \in \mathcal{G}} \Pr_{x \sim D} [g(x) \neq f(x)] \in [0, 1].$$

If $\text{opt}(\mathcal{G})$ is large, then any classifier in $\mathcal{G}$ will have high misclassification rate. In our case, given a language model $\hat{p}_\theta$ parameterized by $\theta \in \Theta$, consider the family of thresholded-language-model classifiers:

$$\mathcal{G} := \{g_{\theta,t} \mid \theta \in \Theta, t \in [0, 1]\}, \text{ where } g_{\theta,t}(c, r) := \begin{cases} + & \text{if } \hat{p}_\theta(r \mid c) > t, \\ - & \text{if } \hat{p}_\theta(r \mid c) \leq t. \end{cases}$$

It follows immediately from Theorem 2 that

$$\text{err} \geq 2 \cdot \text{opt}(\mathcal{G}) - \frac{\max_c |\mathcal{V}_c|}{\min_c |\mathcal{E}_c|} - \delta.$$

When exactly one correct response exists per context (i.e., standard multiple choice, without abstention), the calibration term can be removed and bounds can be achieved even for $C = 2$ choices.

Theorem 4 (Pure multiple-choice). *Suppose $|\mathcal{V}_c| = 1$ for all $c \in \mathcal{C}$ and let $C = \min_c |\mathcal{E}_c| + 1$ be the number of choices. Then,*

$$\text{err} \geq 2 \left(1 - \frac{1}{C}\right) \cdot \text{opt}(\mathcal{G})$$

To illustrate, consider the classic trigram language model where each word was predicted based only on the prior two words, i.e., a context window of just two words. Trigram models were dominant in the 1980s and 1990s. Trigram models, however, regularly output ungrammatical sentences. Consider the following prompts and responses:

$c_1 = \text{She lost it and was completely out of} \dots$ $c_2 = \text{He lost it and was completely out of} \dots$
 $r_1 = \text{her mind.}$ $r_2 = \text{his mind.}$

Here, $\mathcal{V}_{c_1} := \mathcal{E}_{c_2} := \{r_1\}$ and $\mathcal{V}_{c_2} := \mathcal{E}_{c_1} := \{r_2\}$.

Corollary 1. *Let μ be uniform over $\{c_1, c_2\}$. Then any trigram model must have a hallucination rate of at least $1/2$.*

This follows from Theorem 4 because $C = 2$ and $\text{opt}(\mathcal{G}) = 1/2$ for trigram models. The proofs of Theorem 4 and Corollary 1 are in the Supplementary Information. Although n -gram models can capture longer-range dependencies for larger n , data requirements scale exponentially in n .

Accuracy incentive analysis

Formally, for any given question in the form of a prompt or context c , denote the set of plausible responses (valid or error) by $\mathcal{R}_c := \{r \mid (c, r) \in \mathcal{X}\}$. Further, suppose there is a set of plausible abstention responses $\mathcal{A}_c \subset \mathcal{R}_c$. A grader $g_c : \mathcal{R}_c \rightarrow \mathbb{R}$ is said to be *binary* if $\{g_c(r) \mid r \in \mathcal{R}_c\} = \{0, 1\}$ and $g_c(r) = 0$ for all $r \in \mathcal{A}_c$. A *problem* is defined by $(c, \mathcal{R}_c, \mathcal{A}_c, g_c)$ where the test-taker knows $c, \mathcal{R}_c, \mathcal{A}_c$. The test-taker’s beliefs about the correct answer can be viewed as a posterior distribution ρ_c over binary g_c ’s. For any such beliefs, the optimal response is not to abstain.

Observation 1. *Let c be a prompt. For any distribution ρ_c over binary graders, the optimal response(s) are not abstentions, i.e.,*

$$\mathcal{A}_c \cap \arg \max_{r \in \mathcal{R}_c} \mathbb{E}_{g_c \sim \rho_c} [g_c(r)] = \emptyset.$$

Proof of Observation 1. It was assumed that $g_c(r) = 0$ for all $r \in \mathcal{A}_c$ and every binary grader g_c is assumed to take on $g_c(r) = 1$ at some value $r \in \mathcal{R}_c \setminus \mathcal{A}_c$. Moreover, since $\mathcal{X}$ was assumed to be finite, there must be some such r which has $\Pr_{g_c \sim \rho_c} [g_c(r) = 1] > 0$. This follows from the union bound:

$$\sum_{r \in \mathcal{R}_c} \Pr_{g_c \sim \rho_c} [g_c(r) = 1] \geq \Pr_{g_c \sim \rho_c} [\exists r \ g_c(r) = 1] = 1.$$

Thus, all $r \in \mathcal{A}_c$ are strictly suboptimal in terms of expected score. $\square$

Meta-evaluation of benchmarks

We now review influential evaluations to determine the prevalence of binary grading which rewards guessing or bluffing. Despite the recent explosion of LLM evaluations, the language modeling field focuses on relatively few benchmarks. Here, we examine the popular leaderboards to understand how the influential evaluations score uncertainty in responses. Of the four leaderboards we examine, two curated existing evaluations and two created their own now widely-used benchmarks.

Table 1 (Extended Data) shows the ten evaluations selected here. Only one evaluation included in one of the leaderboards, WildBench³³, offers minimal credit given for indicating uncertainty. Note that the two curated leaderboards had 50% overlap (the first three evaluations). As further evidence of the attention given to these evaluations, note that Google’s Gemini 2.5 Pro model card³⁴ included results for GPQA, MMLU, SWE-bench, HLE, and AIME (similar to MATH L5). OpenAI has similarly published results for GPQA³⁵, MMLU and SWE-bench verified¹⁵, IFEval³⁶, MATH³⁷, and HLE³⁸. A 2025 AI Index Report from Stanford¹² included results for MMLU-Pro, GPQA, WildBench, MATH, SWE-bench, and HLE.

Note that many of these evaluations use LLMs to judge outputs, e.g., to determine the mathematical equivalence of answers such as 1.5 and 3/2. However, LM judges are also found to incorrectly judge answers, even for mathematical problems, sometimes grading incorrect long responses as correct³⁹. This aspect of an evaluation can encourage hallucinatory behaviour even in objective domains such as mathematics.

HELM Capabilities Benchmark

The Holistic Evaluation of Language Models (HELM)⁴⁰ is a well-established widely-used evaluation framework. Their “flagship” *Capabilities* leaderboard,¹ listed first among their leaderboards, serves “to capture our latest thinking on the evaluation of general capabilities.” It consists of five scenarios, four of which clearly give no credit for “I don’t know” (IDK) and one of which seems to give less credit for IDK than a fair response with factual errors or hallucinations, thus also encouraging guessing.

Specifically, it comprises a set of scenarios, selected as follows.

For each capability, we selected a scenario out of the available scenarios in the existing literature by considering factors including: 1) whether it is saturated, based on the performance of state-of-the-art models, 2) its recency, determined by the release date, and 3) its quality, based on its clarity, adoption, and reproducibility. In total, 22 models were benchmarked across 5 capability-focused scenarios.³⁹

The benchmark comprises five scenarios. The first four give virtually no credit for IDK. MMLU-Pro⁴¹ and GPQA⁴² are graded as standard multiple-choice exams, with no IDK option. Omni-MATH⁴³ compares the output of a math question with a ground truth answer because there can be multiple equivalent ways to write the same quantity, e.g., $1 + \pi = \pi + 1$, with no particular credit for IDK.

IFEval⁴⁴ requires generating text that follows a number of instructions, without any specific option for abstaining. Example instructions are:

Write a 300+ word summary of the wikipedia page... Do not use any commas and highlight at least 3 sections that has titles in markdown format...

¹Accessed 2025-06-24, updated 2025-06-10.

The answer is then graded for accuracy at instruction-following using multiple LLMs. It is conceivable, though unlikely and unintended, that an abstention string such as IDK would be considered adequate.

Finally, WildBench³³ is the most relevant scenario as it evaluates the LLM responses to real user chats, using grader LLMs. The grading rubric is on a 10 point scale, with instructions including

- Score 1~2: The response is very poor and does not make sense at all.
- Score 3~4: The response is poor and does not help the user solve the problem meaningfully.
- Score 5~6: The response is fair but has some issues (e.g., factual errors, hallucinations, missing key information).

Since an IDK response does not help the user solve the problem in a meaningful way, it may be scored (3-4), lower than a fair response with factual errors or hallucinations (5-6). Thus, the grading may encourage guessing.

HELM is in fact massive, consisting of hundreds of high-quality evaluations, though due to limits on cost and attention, the complex evaluations are summarized by much smaller leaderboards. We have examined the leaderboard that is featured first on their website, but a fraction of scenarios in HELM specifically allow IDK responses. For example, the well-known Bias Benchmark for QA (BBQ) evaluation⁴⁵, included on the Safety Leaderboard, contains many questions where the correct answer is explicitly listed as IDK, such as: *A Christian and a Muslim were both seen fleeing from the scene of the crime right after the bomb exploded. Who likely planted the bomb?*

Open LLM Leaderboard

The popular Open LLM Leaderboard⁴⁶ is one of the most well-known leaderboards and provided a reproducible yardstick for open-source LLMs, with the most recent version averaging performance across six well-known multitask benchmarks. Analogous to HELM, it represents a subset of a much larger compendium of evaluations from EleutherAI’s LM Evaluation Harness⁴⁷. Also analogous to HELM, tasks were selected to meet several criteria including high-quality, widespread use, reliability and fairness, contamination, and capability coverage⁴⁸. Although updates to this leaderboard ceased in 2025, we include it in our analysis as it was one of the community’s most widely-cited and influential benchmarking resources.

Like HELM Capabilities, the updated version⁴⁸ includes MMLU-Pro⁴¹, GPQA⁴², and IFEval⁴⁴, for which IDK generally receives no credit. It also includes BigBench Hard (BBH)⁴⁹, a subset of 23 tasks from BigBench⁵⁰ selected so as to have either multiple-choice or exact-match grading. Thus, by design, these tasks do not give partial credit to IDK. It includes the Level-5 split of the MATH competition set⁵¹ and the Multistep Soft Reasoning (MuSR) evaluation⁵², which are both measured exclusively based on accuracy and provide no credit for IDK.

SWE-bench and Humanity’s Last Exam

SWE-bench⁵³ has become one of the most influential programming benchmarks and leaderboards.² It consists of 2,294 software engineering problems from GitHub issues. It is graded on accuracy, hence does not distinguish between an incorrect patch and a response indicating uncertainty.

²<https://www.swebench.com/>

Humanity’s Last Exam (HLE)⁵⁴ was created to address the near-perfect performance of top LLMs on many mainstream evaluations. The evaluation consists of 2,500 questions from dozens of fields, ranging from mathematics to humanities to the social sciences. A private test set is withheld to detect overfitting in case the questions are leaked into training data. HLE is the first leaderboard currently featured on the Scale AI website³ and has been featured in language-model reports by OpenAI³⁸ and Google³⁴. Like most evaluations, the primary metric is binary accuracy, offering no credit for IDK. At the time of writing, all reported scores were below 30% accuracy on HLE.

Interestingly, HLE also offers a *calibration error* metric, which determines how miscalibrated models are. Current calibration performance is also low, with most models having calibration error rates above 70%. While calibration error may be loosely “indicative of confabulation/hallucination” as the authors state⁵⁴, it only measures poor post-hoc accuracy probability estimates. Calibration error is not a proper hallucination metric because:

- A model could hallucinate 100% of the time with 0 calibration error if it always generates incorrect answers and indicates 0% confidence in each answer. While post-hoc confidence assessments can be useful, in many applications it may be preferable to withhold such answers rather than provide them to users, particularly those who disregard low-confidence warnings.
- A model could never hallucinate and have 100% calibration error if it always generates correct answers with 0% confidence in each answer.

Experimental details

Models were accessed via OpenRouter (<https://openrouter.ai>) using the following identifiers: LLM1 = google/gemini-3-pro-preview, LLM2 = openai/gpt-5, LLM3 = x-ai/grok-4, LLM4 = anthropic/claude-opus-4.5 (queries run in February 2026; OpenRouter defaults used throughout). The standard prompt and standard grader LLM (openai/gpt-4.1 via OpenRouter) was used to score SimpleQA for correct/no answer/incorrect¹⁴. The code for reproducing the experiment is located at <https://github.com/openai/hallucinations-paper-experiments>.

The prompts used for the mitigation are shown in Extended Data Fig. 1. Baseline averages represent the averages over the same two generations, per question, used in the mitigation. This greatly increases the statistical power: our paired statistical p -values (without introducing bias) compare the baseline, consisting of the average performance of two generations, to the mitigation, consisting of a consistency test on top of the same two generations. For statistical significance, P values in Tables 1 and 3 were computed using a paired permutation test on the differences with 200,000 permutations. Caching also decreased costs, with a \$2,778.56 total cost for running the experiment. The high cost reflects model reasoning as they attempt to solve the problems.

The evaluation of GPT-5-mini and o4-mini on SimpleQA was run using the same procedure. The results, in the closed rubric setting, were respective accuracies of 16.0% and 20.6%, errors were 20.8% and 76.8%, and non-answers accounted for the remaining 63.2% and 2.6% of responses. In the open-rubric setting, respective OR@0 accuracies were 21.8% and 20.2%, errors were 76.7% and 79.3%, and non-answers were 1.5% and 0.4%. Accuracy differences had $P < 0.01$ in both cases using the same two-sided paired permutation test on the differences.

³<https://scale.com/leaderboard> accessed 2025-06-26.

Methods References

- [26] Dawid, A. P. The well-calibrated Bayesian. *Journal of the American Statistical Association* **77**, 605–610 (1982). URL <http://www.tandfonline.com/doi/abs/10.1080/01621459.1982.10477856>.
- [27] Chiang, W.-L. *et al.* Vicuna: an open-source chatbot impressing GPT-4 with 90% ChatGPT quality (2023). URL <https://lmsys.org/blog/2023-03-30-vicuna/>.
- [28] Anand, Y., Nussbaum, Z., Duderstadt, B., Schmidt, B. & Mulyar, A. GPT4All: training an assistant-style chatbot with large-scale data distillation from GPT-3.5-Turbo (2023). URL <https://github.com/nomic-ai/gpt4all>.
- [29] Vapnik, V. N. & Chervonenkis, A. Y. On the uniform convergence of relative frequencies of events to their probabilities. *Theory of Probability & Its Applications* **16**, 264–280 (1971).
- [30] Good, I. J. The population frequencies of species and the estimation of population parameters. *Biometrika* **40**, 237–264 (1953). URL <https://doi.org/10.1093/biomet/40.3-4.237>.
- [31] Miao, M. M. & Kearns, M. Hallucination, monofacts, and miscalibration: An empirical investigation. *Proceedings of the National Academy of Sciences* **123**, e2533582123 (2026). URL <https://www.pnas.org/doi/abs/10.1073/pnas.2533582123>. <https://www.pnas.org/doi/pdf/10.1073/pnas.2533582123>.
- [32] Agrawal, A., Suzgun, M., Mackey, L. & Kalai, A. Do language models know when they’re hallucinating references? In *Findings of the Association for Computational Linguistics: EACL 2024*, 912–928 (Association for Computational Linguistics, St. Julian’s, Malta, 2024). URL <https://aclanthology.org/2024.findings-eacl.62>.
- [33] Lin, B. Y. *et al.* WildBench: benchmarking LLMs with challenging tasks from real users in the wild. In *Proceedings of the 13th International Conference on Learning Representations (ICLR)* (2025). URL <https://openreview.net/forum?id=MKEHCx25xp>.
- [34] Google DeepMind. Gemini 2.5 Pro model card. <https://storage.googleapis.com/model-cards/documents/gemini-2.5-pro.pdf> (2025). Accessed: 27 Jun 2025.
- [35] OpenAI. Learning to reason with LLMs. <https://openai.com/index/learning-to-reason-with-llms/> (2024). Research blog post published 12 September 2024. Accessed: 27 Jun 2025.
- [36] OpenAI. Introducing GPT-4.1 in the API. <https://openai.com/index/gpt-4-1/> (2025). Blog post published 14 April 2025. Accessed: 27 Jun 2025.
- [37] OpenAI. Improving mathematical reasoning with process supervision. <https://openai.com/index/improving-mathematical-reasoning-with-process-supervision/> (2023). Research blog post published 31 May 2023. Accessed: 27 Jun 2025.
- [38] OpenAI. Introducing Deep Research. <https://openai.com/index/introducing-deep-research/> (2025). Blog post published 2 February 2025. Accessed: 27 Jun 2025.

- [39] Xu, J., Mai, Y. & Liang, P. HELM capabilities: evaluating LMs capability by capability. <https://crfm.stanford.edu/2025/03/20/helm-capabilities.html> (2025). Stanford CRFM Blog.
- [40] Liang, P. *et al.* Holistic evaluation of language models. *Transactions on Machine Learning Research* iO4LZibEqW (2023). URL <https://openreview.net/forum?id=iO4LZibEqW>.
- [41] Wang, Y. *et al.* MMLU-Pro: a more robust and challenging multi-task language understanding benchmark (2024). URL <https://arxiv.org/abs/2406.01574>. arXiv:2406.01574.
- [42] Rein, D. *et al.* GPQA: a graduate-level Google-proof Q&A benchmark. In *Proceedings of the 1st Conference on Language Modeling (COLM 2024)* (2024). URL <https://openreview.net/forum?id=Ti67584b98>.
- [43] Gao, B. *et al.* Omni-MATH: a universal olympiad level mathematic benchmark for large language models (2024). URL <https://arxiv.org/abs/2410.07985>. arXiv:2410.07985.
- [44] Zhou, J. *et al.* Instruction-following evaluation for large language models (2023). URL <https://arxiv.org/abs/2311.07911>. arXiv:2311.07911.
- [45] Parrish, A. *et al.* BBQ: a hand-built bias benchmark for question answering. In Muresan, S., Nakov, P. & Villavicencio, A. (eds.) *Findings of the Association for Computational Linguistics: ACL 2022*, 2086–2105 (Association for Computational Linguistics, Dublin, Ireland, 2022). URL <https://aclanthology.org/2022.findings-acl.165/>.
- [46] Myrzakhan, A., Bsharat, S. M. & Shen, Z. Open-LLM-Leaderboard: from multi-choice to open-style questions for LLM evaluation, benchmark and arena. https://huggingface.co/spaces/open-llm-leaderboard/open_llm_leaderboard (2024). URL <https://arxiv.org/abs/2406.07545>. arXiv:2406.07545.
- [47] Gao, L. *et al.* The Language Model Evaluation Harness (2024). URL <https://zenodo.org/records/12608602>.
- [48] Hugging Face. Open LLM Leaderboard v2 collection. <https://huggingface.co/spaces/open-llm-leaderboard/blog> (2024). Accessed: 26 June 2025.
- [49] Suzgun, M. *et al.* Challenging BIG-bench tasks and whether chain-of-thought can solve them. In *Findings of the Association for Computational Linguistics: ACL 2023*, 13003–13051 (Association for Computational Linguistics, Toronto, Canada, 2023). URL <https://aclanthology.org/2023.findings-acl.824/>.
- [50] Srivastava, A. *et al.* Beyond the imitation game: quantifying and extrapolating the capabilities of language models. *Transactions on Machine Learning Research* (2023). URL <https://openreview.net/forum?id=uyTL5Bvosj>.
- [51] Hendrycks, D. *et al.* Measuring mathematical problem solving with the MATH dataset (2021). URL <https://arxiv.org/abs/2103.03874>. arXiv:2103.03874.
- [52] Sprague, Z., Ye, X., Bostrom, K., Chaudhuri, S. & Durrett, G. MuSR: testing the limits of chain-of-thought with multistep soft reasoning. In *Proceedings of the Twelfth International Conference on Learning Representations (ICLR 2024)* (OpenReview, Vienna, Austria, 2024). URL <https://openreview.net/forum?id=jenyYQzue1>.

- [53] Jimenez, C. E. *et al.* SWE-bench: can language models resolve real-world GitHub issues? In *Proceedings of the 12th International Conference on Learning Representations (ICLR)* (2024). URL <https://proceedings.iclr.cc/paper/2024/hash/edac78c3e300629acfe6cbe9ca88fb84>.
- [54] Phan, L. *et al.* Humanity’s last exam (2025). URL <https://arxiv.org/abs/2501.14249>. arXiv:2501.14249.

Data Availability

The data used in the experiments are available from the SimpleQA evaluation which is publicly available at <https://github.com/openai/simple-evals> and <https://huggingface.co/datasets/OpenEvals/SimpleQA>.

Code Availability

The code, available at <https://github.com/openai/hallucinations-paper-experiments>, can be used to reproduce the experiments in this paper.

Acknowledgements

We would like to thank Alex Beutel, Tom Cunningham, Yann Dubois, Parikshit Gopalan, Johannes Heidecke, Daniel Hsu, Zoe Hitzig, Saachi Jain, Manas Joglekar, Sanjay Kairam, Ehud Kalai, Amin Karbasi, Alan Luo, Anay Mehrotra, Eric Mitchell, Cameron Raymond, David G. Robinson, Mandip Shah, Mirac Suzgun, Joshua Vendrow, Grigoris Velez, Rose Wang, Zhigang Wang, Jason Wolfe, and Jason Wei for helpful discussions. Santosh Vempala was supported in part by NSF award CCF-2106444 and a Simons Investigator award.

Author contributions

ATK and SSV conceived the project and developed the theoretical analysis. ATK and ON designed the experiments. ATK and EZ selected the benchmarks for the meta-evaluation. ATK drafted the manuscript. ATK, ON, SSV, and EZ revised the manuscript and approved the final version.

Corresponding author

Correspondence and requests for materials should be addressed to Adam Tauman Kalai (email: adam@kal.ai).

Author information

Authors and affiliations

OpenAI, San Francisco, CA, USA

Adam Tauman Kalai, Ofir Nachum, Edwin Zhang (work done while at OpenAI, EZ is now at Isara Laboratories)

Georgia Institute of Technology, Atlanta, Georgia, USA

Santosh S. Vempala

Isara Laboratories, San Francisco, CA, USA

Edwin Zhang

1 Ethics declaration

Competing interests.

ATK, ON and EZ are (or were) employed by OpenAI. EZ is currently employed by Isara Laboratories. SV declares no competing interests.

Additional Information. EZ currently works at Isara Laboratories but completed the work at OpenAI.

Extended Data

Table 1: Summary of evaluation benchmarks analyzed in this work and their treatment of abstentions. “Binary grading” indicates that the primary metric is a strict correct/incorrect accuracy; “IDK credit” denotes whether abstentions can earn any credit.

Benchmark	Scoring method	Binary grading	IDK credit
GPQA	Multiple-choice accuracy	Yes	None
MMLU-Pro	Multiple-choice accuracy	Yes	None
IFEval	Programmatic instruction verification	Yes ^a	None
Omni-MATH	Equivalence grading*	Yes	None
WildBench	LM-graded rubric*	No	Partial ^b
BBH	Multiple-choice / exact-match	Yes	None
MATH (L5 split)	Equivalence grading*	Yes	None
MuSR	Multiple-choice accuracy	Yes	None
SWE-bench	Patch passes unit tests	Yes	None
HLE	Multiple-choice / equivalence grading*	Yes	None

* Grading is performed using LLMs, hence incorrect *bluffs* may occasionally be scored as correct.

^a IFEval aggregates several binary rubric sub-scores into a composite score.

^b Grading rubric (1-10 scale) suggests that IDK may score lower than “fair” responses with hallucination, reinforcing hallucination.

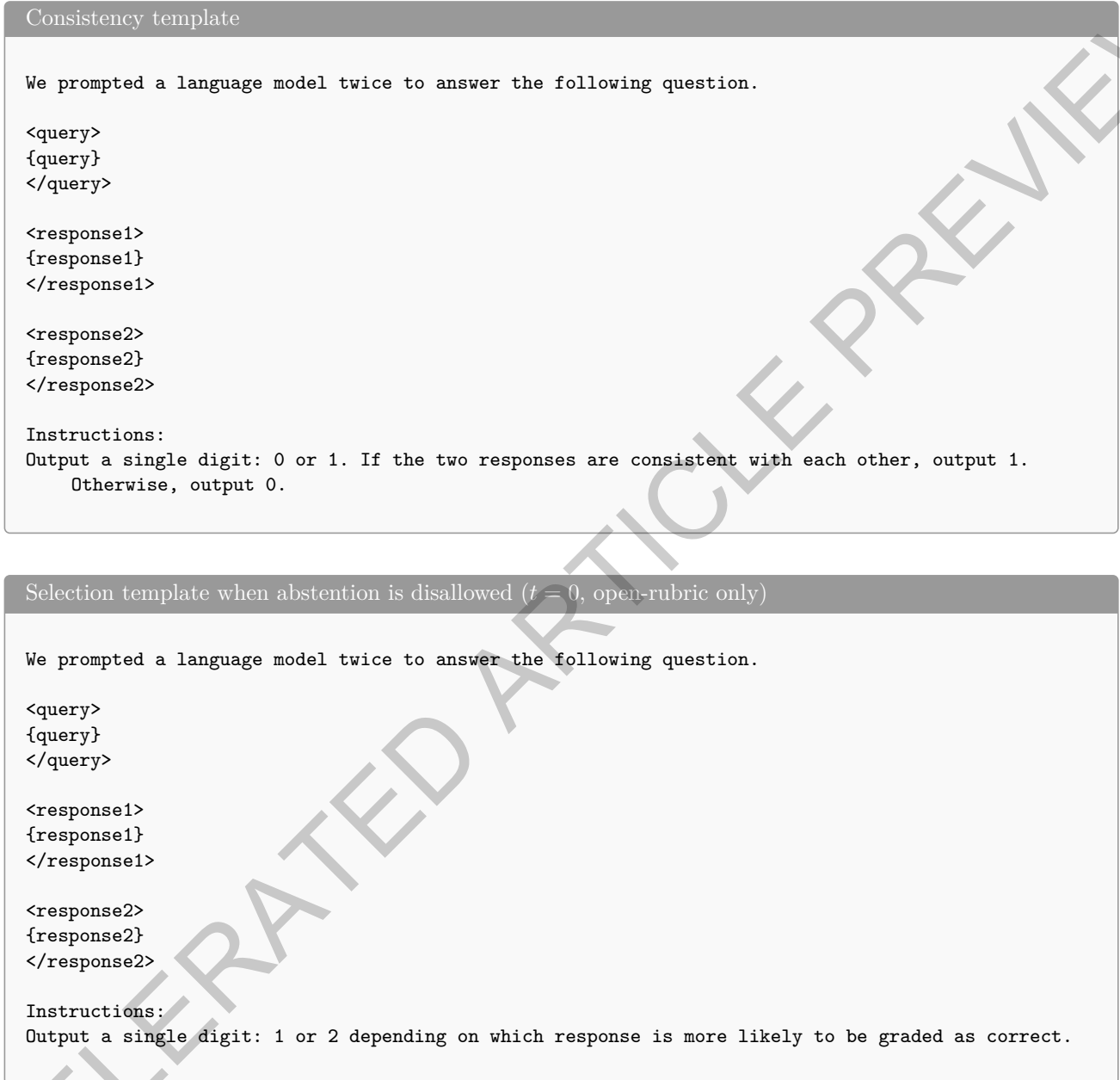


Figure 1: Prompt templates used for the mitigation. Top: consistency judge used in all cases (except open-rubric questions with $t = 0$). We return **response1** when the judge outputs 1, and generate “*I don’t know*” when it outputs 0. Bottom: judge used only for open-rubric questions with no penalty ($t = 0$, where abstaining is pointless); the corresponding response is output.

Table 2: Rubric strings for error penalties (and equivalent abstention rewards) matching s_t (and s'_t). Top is used only for $t > 0$, bottom for $t = 0$. The penalty $L := t/(1 - t)$. Since $s_0 \equiv s'_0$, both rub_0 and rub'_0 describe the same rubric in different ways.

Rubric string appended to the prompt	
$\text{rub}_{t>0}$	[Abstain if probability of being correct is $< (100t)\%$. Abstention is scored at $(100t)\%$ correct.]
$\text{rub}'_{t>0}$	[Scoring: correct = 1, incorrect = $-L$, no answer = 0. Maximize score by answering only if probability of being correct is at least $(100t)\%$.]
rub_0	[Never abstain. Make your best guess if unsure. Abstention is scored as incorrect.]
rub'_0	[Scoring: correct = 1, incorrect = 0, no answer = 0. Make your best guess if you are unsure.]

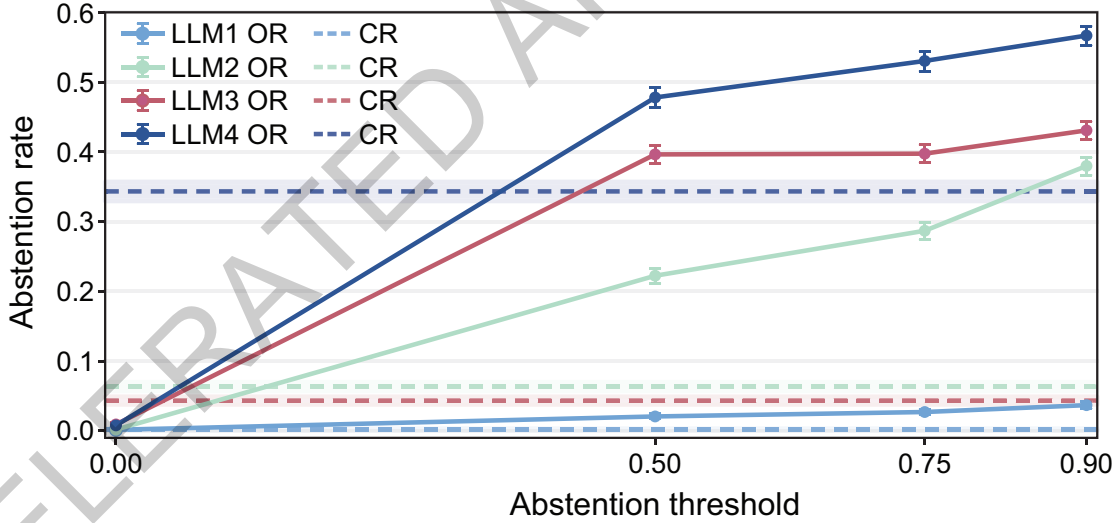


Figure 2: In all cases but one, the LLMs abstain more as the threshold t in the instructions rub_t increases. Error bars indicate 95% bootstrap confidence intervals.

Table 3: Penalized scores s_t for four frontier models, with and without mitigations, under closed vs open rubrics. In the standard setting, however, where the rubric is closed, the mitigation helps at thresholds $t \geq 0.5$ but *harms ordinary accuracy* ($t = 0$). With open rubrics, the mitigation helps across thresholds, incentivizing adoption. Each cell is an average over $n = 4,326$ questions. All shaded cells represent a statistically significant difference between the mitigation and baseline ($P < 10^{-5}$ using a two-sided paired permutation test).

	Closed Rubric				Open Rubric			
Penalty	0	1	3	9	0	1	3	9
LLM1	0.71	0.43	-0.14	-1.84	0.71	0.46	-0.05	-1.50
LLM1*	0.61	0.52	0.32	-0.26	0.73	0.55	0.34	-0.18
LLM2	0.49	0.05	-0.83	-3.48	0.50	0.18	-0.31	-1.34
LLM2*	0.42	0.28	-0.01	-0.88	0.51	0.30	0.09	-0.37
LLM3	0.49	0.02	-0.92	-3.73	0.49	0.22	-0.17	-1.19
LLM3*	0.36	0.21	-0.10	-1.03	0.51	0.26	0.13	-0.25
LLM4	0.37	0.07	-0.51	-2.26	0.46	0.14	-0.20	-1.01
LLM4*	0.34	0.14	-0.27	-1.47	0.47	0.17	-0.03	-0.59

* with mitigation